Radical Expressions Name:__________

5.1

1. Square Roots

Definition. The *square root* of a number x is any number that, when squared, equals x. For example, the square roots of 9 are 3 and -3, because $3^2 = 9$ and $(-3)^2 = 9$.
The symbol $\sqrt{\ }$ is used for the *principal* (positive) square root. For example, $\sqrt{9} = 3$ and $-\sqrt{9} = -3$.

A. Simplify each.

__________ 1. $\sqrt{49}$

__________ 2. $\sqrt{25}$

__________ 3. $\sqrt{81}$

__________ 4. $\sqrt{100}$

__________ 5. $\sqrt{36}$

__________ 6. $\sqrt{1}$

__________ 7. $\sqrt{16}$

__________ 8. $\sqrt{0}$

__________ 9. $\sqrt{64}$

__________ 10. $\sqrt{10000}$

B. Use the definition of a square root.

If $\sqrt{x}$ is a square root of x, then $(\sqrt{x})^2 = x$. More generally, if $a > 0$, then $(\sqrt{a})^2 = a$ and $\sqrt{a} \cdot \sqrt{a} = a$.

Compute each expression.

__________ 11. $(\sqrt{16})^2$

__________ 12. $(\sqrt{25})^2$

__________ 13. $(\sqrt{100})^2$

__________ 14. $(\sqrt{225})^2$

__________ 15. $(\sqrt{49})^2$

__________ 16. $(\sqrt{900})^2$

__________ 17. $\sqrt{4} \cdot \sqrt{4}$

__________ 18. $\sqrt{9} \cdot \sqrt{9}$

__________ 19. $\sqrt{16} \cdot \sqrt{16}$

__________ 20. $\sqrt{49} \cdot \sqrt{49}$

__________ 21. $\sqrt{225} \cdot \sqrt{225}$

__________ 22. $\sqrt{196} \cdot \sqrt{196}$

__________ 23. $(\sqrt{169})^2$

__________ 24. $\sqrt{1} \cdot \sqrt{1}$

__________ 25. $\sqrt{0} \cdot \sqrt{0}$

2. More Radicals

The expression $\sqrt[m]{a}$ is called a *radical expression* (or just a *radical*) of the mth degree. Here:

- $\sqrt[m]{}$ is the *radical sign*,
- a is the *radicand*,
- m is the *index* (an integer with $m \geq 2$).

The number $\sqrt[m]{a}$ is the number that, when raised to the mth power, equals a. Symbolically, $(\sqrt[m]{a})^m = a$.

A. Exponent form

Recall that $\sqrt[m]{a^p} = a^{\frac{p}{m}}$ when $a > 0$.

Write each in exponent form.

__________ 1. $\sqrt[3]{x^2}$

__________ 2. $\sqrt[4]{3^2}$

__________ 3. $6\sqrt[7]{5^3}$

__________ 4. $\sqrt[4]{3}$

__________ 5. $\sqrt[5]{y^4}$

__________ 6. $\sqrt[6]{4x}$

B. Radical form

Write each in radical form.

__________ 7. $x^{\frac{1}{4}}$

__________ 8. $3^{\frac{2}{5}}$

__________ 9. $9^{\frac{1}{2}}$

__________ 10. $4^{\frac{3}{5}}$

__________ 11. $12^{\frac{1}{4}}$

__________ 12. $6^{\frac{3}{2}}$

C. Simplify completely

__________ 13. $\sqrt[3]{16}$

__________ 14. $\sqrt[4]{81}$

__________ 15. $\sqrt[3]{64}$

__________ 16. $(\sqrt{16})^2$

__________ 17. $\sqrt[3]{125} \cdot \sqrt[3]{5}$

__________ 18. $\sqrt[4]{16x^4}$

__________ 19. $\sqrt[3]{27y^6}$

__________ 20. $\sqrt[5]{32x^{10}}$

3. Radical Drill Sheet

Recall that $a^{\frac{1}{2}} = \sqrt{a}$ and $\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$ for $a, b \geq 0$. These rules allow us to simplify radicals by factoring the radicand.

Simplify each radical.

______ 1. $\sqrt{13 \cdot 13 \cdot 13}$

______ 2. $\sqrt{5 \cdot 5 \cdot 5 \cdot 5 \cdot 5}$

______ 3. $\sqrt{3 \cdot 3}$

______ 4. $\sqrt{3 \cdot 3 \cdot 3 \cdot 3}$

______ 5. $\sqrt{2 \cdot 3 \cdot 3}$

______ 6. $\sqrt{3 \cdot 3 \cdot 3 \cdot 5}$

______ 7. $\sqrt{2 \cdot 3 \cdot 3 \cdot 5}$

______ 8. $\sqrt{2 \cdot 3 \cdot 5 \cdot 5}$

______ 9. $\sqrt{2 \cdot 3 \cdot 5 \cdot 5 \cdot 5}$

______ 10. $\sqrt{2 \cdot 2 \cdot 2 \cdot 3 \cdot 3 \cdot 3}$

______ 11. $\sqrt{2 \cdot 3 \cdot 5}$

______ 12. $\sqrt{98}$

______ 13. $\sqrt{27}$

______ 14. $\sqrt{50}$

______ 15. $\sqrt{12}$

______ 16. $\sqrt{8}$

______ 17. $\sqrt{100}$

______ 18. $\sqrt{20}$

______ 19. $\sqrt{40}$

______ 20. $\sqrt{45}$

______ 21. $\sqrt{36}$

______ 22. $\sqrt{44}$

______ 23. $\sqrt{25}$

______ 24. $\sqrt{75}$

______ 25. $\sqrt{72}$

______ 26. $\sqrt{81}$

______ 27. $\sqrt{32}$

______ 28. $\sqrt{35}$

______ 29. $\sqrt{54}$

______ 30. $\sqrt{48}$

______ 31. $\sqrt{24}$

______ 32. $\sqrt{60}$

______ 33. $\sqrt{4}$

______ 34. $\sqrt{72}$

______ 35. $\sqrt{16}$

______ 36. $\sqrt{80}$

______ 37. $\sqrt{64}$

______ 38. $\sqrt{10}$

______ 39. $\sqrt{200}$

______ 40. $\sqrt{18}$

4. Simplifying Radical Expressions

A radical expression is in *simplest form* if:

- the number of identical factors of any kind under the radical is fewer than the index,
- no radical appears in a denominator,
- no fraction appears under a radical.

A. Simplify completely (integer radicands).

________ 1. $\sqrt[3]{720}$

________ 2. $\sqrt[3]{24}$

________ 3. $\sqrt[3]{160}$

________ 4. $\sqrt[4]{81}$

________ 5. $\sqrt[4]{625}$

________ 6. $\sqrt[3]{160}$

________ 7. $\sqrt[3]{128}$

________ 8. $\sqrt[4]{256}$

________ 9. $\sqrt[3]{250}$

________ 10. $\sqrt[3]{216}$

B. Rationalizing denominators

To *rationalize the denominator* of a fraction like $\frac{3}{\sqrt{2}}$, multiply the numerator and denominator by a radical that clears the radical from the denominator:

$$\frac{3}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{3\sqrt{2}}{2}.$$

Rewrite each expression so that no radical remains in the denominator.

________ 11. $\frac{3}{\sqrt{7}}$

________ 12. $\frac{7}{\sqrt{5}}$

________ 13. $\frac{3}{\sqrt{2}}$

________ 14. $\frac{6}{\sqrt{2}}$

________ 15. $\frac{3}{\sqrt{6}}$

________ 16. $\frac{6}{\sqrt{8}}$

________ 17. $\frac{10}{\sqrt{10}}$

________ 18. $\frac{\sqrt{5}}{\sqrt{2}}$

__________ 19. $\dfrac{\sqrt{6}}{\sqrt{3}}$

__________ 20. $\dfrac{2}{\sqrt{2}}$

5. Simplifying Radicals with Fractional Radicands

If $a \geq 0$ and $b > 0$, then

$$\sqrt{\frac{a}{b}} = \left(\frac{a}{b}\right)^{\frac{1}{2}} = \frac{a^{\frac{1}{2}}}{b^{\frac{1}{2}}} = \frac{\sqrt{a}}{\sqrt{b}}.$$

We often use this rule together with simplification and rationalizing denominators.

A. Simplify each radical.

__________ 1. $\sqrt{\dfrac{9}{20}}$

__________ 2. $\sqrt{\dfrac{5}{12}}$

__________ 3. $\sqrt{\dfrac{25}{98}}$

__________ 4. $\sqrt{\dfrac{9}{8}}$

__________ 5. $\sqrt{\dfrac{12}{27}}$

__________ 6. $\sqrt{\dfrac{25}{72}}$

__________ 7. $\sqrt{\dfrac{20}{75}}$

__________ 8. $\sqrt{\dfrac{5}{54}}$

__________ 9. $\sqrt{\dfrac{25}{24}}$

__________ 10. $\sqrt{\dfrac{4}{45}}$

B. Combine like radicals and simplify completely.

__________ 11. $3\sqrt{12} + 5\sqrt{18} - 2\sqrt{75}$

__________ 12. $2\sqrt{12} - 3\sqrt{48} + 5\sqrt{27}$

__________ 13. $5\sqrt{27} - 8\sqrt{12} + 3\sqrt{75}$

__________ 14. $\dfrac{5}{\sqrt{20}}$

__________ 15. $\sqrt{720}$